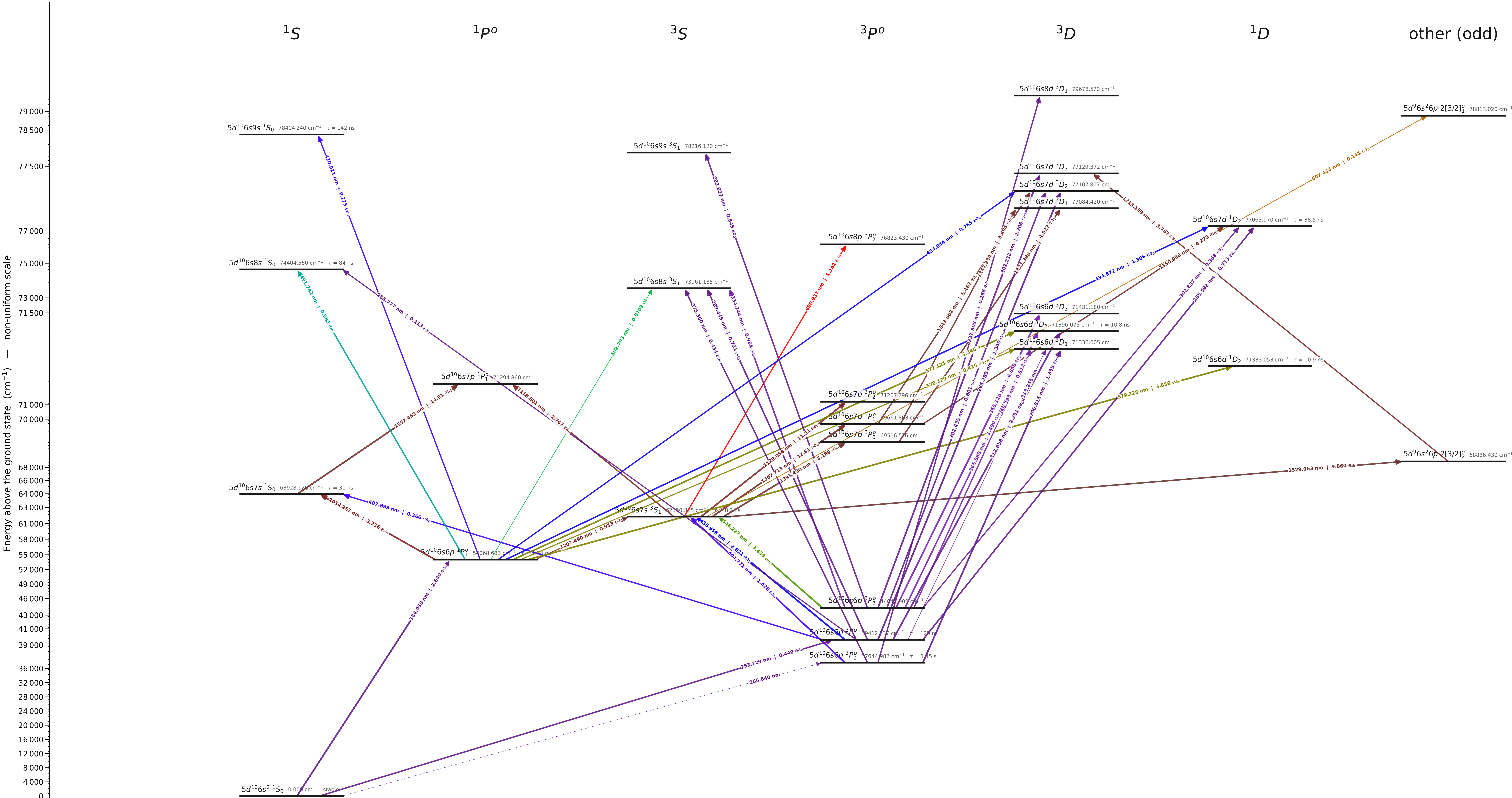
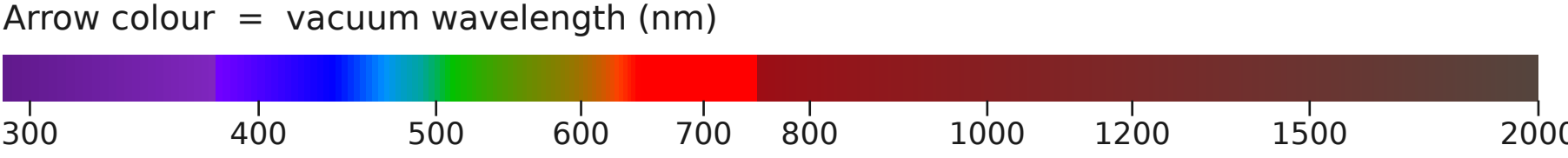


# Mercury

Hg (Hg I). Strongest classified transitions below 2 μm (NIST): 27 levels, 47 lines.

Hg <sup>+</sup> ionisation limit 84184.15 cm<sup>-1</sup> = 10.43750 eV (118.787 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element  $|(U||er||J')|$  in  $e a_0$   
Matrix elements are measured or NIST values; \* = high-accuracy theory, ≈ = model calculation.  
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation,  $n = 70$ .  
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



## Key transitions

| Transition                                  | λ vac (nm) | λ air (nm) | ν (THz)     | $ (U  er  J') $ (ea <sub>0</sub> ) | A (s <sup>-1</sup> ) | Γ/2π up. (MHz) | branch (%) | Use                                                          |
|---------------------------------------------|------------|------------|-------------|------------------------------------|----------------------|----------------|------------|--------------------------------------------------------------|
| $5d^{10}6s^2\ ^1S_0 - 5d^{10}6s6p\ ^1P_1^o$ | 184.9499   | -          | 1620.9383   | 2.640                              | 7.440e+08            | 118.8          | 99.7       | 184.9 nm resonance line                                      |
| $5d^{10}6s^2\ ^1S_0 - 5d^{10}6s6p\ ^3P_1^o$ | 253.7291   | 253.6528   | 1181.545677 | 0.440                              | 7.992e+06            | 1.272          | 100        | 253.7 nm intercombination line: MOT / cooling, optical pu... |
| $5d^{10}6s^2\ ^1S_0 - 5d^{10}6s6p\ ^3P_0^o$ | 265.6397   | 265.5606   | 1128.5682   | -                                  | 6.920e-01            | 1.101e-07      | 100        | 265.6 nm optical lattice clock transition                    |
| $5d^{10}6s6p\ ^3P_0^o - 5d^{10}6s8s\ ^1S_0$ | 285.7770   | 285.6930   | 1049.0434   | 0.113                              | 1.100e+06            | 1.895          | 9.24       | 285.7 nm                                                     |
| $5d^{10}6s6p\ ^3P_0^o - 5d^{10}6s6d\ ^3D_1$ | 296.8150   | 296.7283   | 1010.0315   | 1.335                              | 4.600e+07            | -              | -          | 296.7 nm                                                     |
| $5d^{10}6s6p\ ^3P_1^o - 5d^{10}6s6d\ ^3D_2$ | 312.6579   | 312.5673   | 958.8513    | 2.231                              | 6.000e+07            | 14.74          | 71.3       | 312.7 nm                                                     |
| $5d^{10}6s6p\ ^3P_1^o - 5d^{10}6s6d\ ^3D_1$ | 313.2462   | 313.1555   | 957.0505    | -                                  | -                    | -              | -          | 313.2 nm, optical pumping out of the 253.7 nm MOT            |
| $5d^{10}6s6p\ ^3P_2^o - 5d^{10}6s6d\ ^3D_3$ | 365.1198   | 365.0158   | 821.0797    | 4.658                              | 1.290e+08            | -              | -          | 365.0 nm                                                     |
| $5d^{10}6s6p\ ^3P_0^o - 5d^{10}6s7s\ ^3S_1$ | 404.7707   | 404.6564   | 740.6476    | 1.426                              | 2.070e+07            | 19.41          | 17         | 404.7 nm                                                     |
| $5d^{10}6s6p\ ^3P_1^o - 5d^{10}6s7s\ ^1S_0$ | 407.8988   | 407.7837   | 734.9677    | 0.366                              | 4.000e+06            | 5.134          | 12.4       | 407.8 nm                                                     |
| $5d^{10}6s6p\ ^3P_1^o - 5d^{10}6s7s\ ^3S_1$ | 435.9561   | 435.8336   | 687.6666    | 2.621                              | 5.600e+07            | 19.41          | 45.9       | 435.8 nm                                                     |
| $5d^{10}6s6p\ ^1P_1^o - 5d^{10}6s8s\ ^1S_0$ | 491.7418   | 491.6045   | 609.6543    | 0.583                              | 5.800e+06            | 1.895          | 48.7       | 491.6 nm                                                     |
| $5d^{10}6s6p\ ^3P_2^o - 5d^{10}6s7s\ ^3S_1$ | 546.2267   | 546.0750   | 548.8425    | 3.439                              | 4.900e+07            | 19.41          | 40.2       | 546.1 nm                                                     |
| $5d^{10}6s6p\ ^1P_1^o - 5d^{10}6s7s\ ^1S_0$ | 1014.2567  | 1013.9787  | 295.5785    | 3.736                              | 2.710e+07            | 5.134          | 84         | 1014 nm                                                      |
| $5d^{10}6s6p\ ^1P_1^o - 5d^{10}6s7s\ ^3S_1$ | 1207.4900  | 1207.1596  | 248.2774    | 0.913                              | 3.200e+05            | 19.41          | 0.26       | 1207 nm intercombination branch of 7s 3S1                    |

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.  
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

## <sup>201</sup>Hg hyperfine structure (I = 3/2)

| Level                  | A (MHz)       | B (MHz) | Shift of each F level from the centre of gravity (MHz)             |
|------------------------|---------------|---------|--------------------------------------------------------------------|
| $5d^{10}6s6p\ ^3P_1^o$ | -5454.569(3)  | -       | F=1/2: +13636.42 F=3/2: +5454.57 F=5/2: -8181.85                   |
| $5d^{10}6s6p\ ^3P_2^o$ | -3352.0292(8) | -       | F=1/2: +15084.13 F=3/2: +10056.09 F=5/2: +1676.02 F=7/2: -10056.09 |
| $5d^{10}6s6p\ ^1P_1^o$ | 1316.000      | -       | F=1/2: -3290.00 F=3/2: -1316.00 F=5/2: +1974.00                    |

Measured A, B constants; sources in the data file.

## Isotope shifts

| Transition                                  | λ vac (nm) | Isotopes | Shift (MHz)    |
|---------------------------------------------|------------|----------|----------------|
| $5d^{10}6s^2\ ^1S_0 - 5d^{10}6s6p\ ^3P_1^o$ | 253.729    | 196-198  | 4179.95(3)     |
| $5d^{10}6s^2\ ^1S_0 - 5d^{10}6s6p\ ^3P_1^o$ | 253.729    | 200-198  | -4805.52(4)    |
| $5d^{10}6s^2\ ^1S_0 - 5d^{10}6s6p\ ^3P_1^o$ | 253.729    | 202-198  | -10101.224(22) |
| $5d^{10}6s^2\ ^1S_0 - 5d^{10}6s6p\ ^3P_1^o$ | 253.729    | 204-198  | -15311.60(4)   |
| $5d^{10}6s^2\ ^1S_0 - 5d^{10}6s6p\ ^3P_1^o$ | 253.729    | 199-198  | -652(6)        |
| $5d^{10}6s^2\ ^1S_0 - 5d^{10}6s6p\ ^3P_1^o$ | 253.729    | 201-198  | -6410(6)       |
| $5d^{10}6s^2\ ^1S_0 - 5d^{10}6s6p\ ^3P_0^o$ | 265.640    | 199-201  | 5729.668(8)    |
| $5d^{10}6s^2\ ^1S_0 - 5d^{10}6s6p\ ^3P_0^o$ | 265.640    | 198-199  | 654.48         |
| $5d^{10}6s6p\ ^3P_1^o - 5d^{10}6s6d\ ^3D_2$ | 312.658    | 196-198  | -289.4(12)     |
| $5d^{10}6s6p\ ^3P_1^o - 5d^{10}6s6d\ ^3D_2$ | 312.658    | 200-198  | 328.0(6)       |
| $5d^{10}6s6p\ ^3P_1^o - 5d^{10}6s6d\ ^3D_2$ | 312.658    | 202-198  | 686.7(6)       |
| $5d^{10}6s6p\ ^3P_2^o - 5d^{10}6s6d\ ^3D_2$ | 312.658    | 204-198  | 1038.3(6)      |
| $5d^{10}6s6p\ ^3P_1^o - 5d^{10}6s6d\ ^3D_1$ | 313.246    | 196-198  | -307.5(10)     |
| $5d^{10}6s6p\ ^3P_2^o - 5d^{10}6s6d\ ^3D_1$ | 313.246    | 200-198  | 347.2(6)       |
| $5d^{10}6s6p\ ^3P_2^o - 5d^{10}6s6d\ ^3D_1$ | 313.246    | 202-198  | 725.5(6)       |
| $5d^{10}6s6p\ ^3P_1^o - 5d^{10}6s6d\ ^3D_1$ | 313.246    | 204-198  | 1097.8(6)      |

Shift = ν(first isotope) − ν(second isotope).